

Tutorial on Diffusion Models for Image Generation

Sanjay Shakkottai
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Motivation: Images can be represented as high-dimensional vector of pixels; a 512×512 image in RGB $\approx$ vector in 800,000 dimensions.

Tasks:

- * Generate a new image 'x'
- * Generate an image consistent with text 'z'
- * Denoise / increase resolution / deblur an image given measurement vector 'y'
-

Observation: These generation tasks can be interpreted as sampling from some (high dimensional) distribution or conditional sampling:

$$x \sim p_0(x), \quad x \in \mathbb{R}^d$$

$\hookrightarrow$ distribution over images

$$x \sim p_0(x|y), \quad y \in \mathbb{R}^m$$

$\hookrightarrow$ measurement

$$x \sim p_0(x|z), \quad z \in \mathbb{R}^d$$

↳ "text embedding"

Approach: Construct a Markov process whose stationary distribution is $p_0(\cdot)$. Initialize the MP with a random (Gaussian) initialization, and evolve the MP. Draw a sample after time T large enough, and this should be "close" to an image.

* Markov process used is a SDE (Stochastic Differential Equation). Construct two SDEs:

- ① An SDE that converts images into noise
- ② A "reverse" SDE that converts noise into images.

Outline of Part 1

1. What is an SDE?
2. How does the pdf of the SDE evolve?
→ Fokker-Planck Equation.

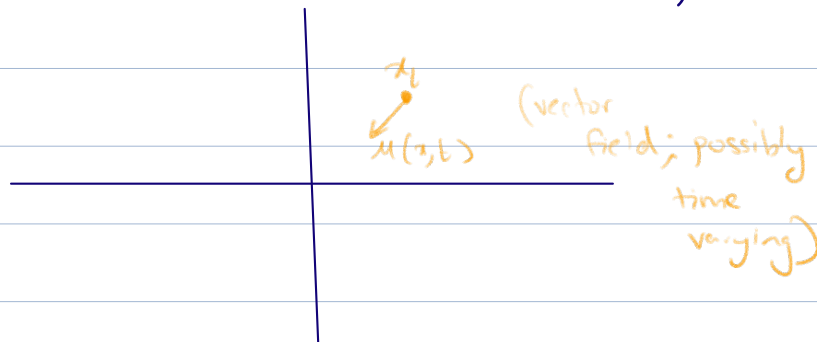
3. The Ornstein-Uhlenbeck (OU) Process
(a specific SDE), with $X \sim \mathcal{N}(\mu, \sigma^2)$.
 4. Reversing the OU process to generate images.
 5. Tweedie's Formula to compute the drift of the reverse process.
 6. Generative flow models using ODEs.
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References (for Part 1) (additional ref in text in appropriate sections)

- [1]. Bernt Oksendal, Stochastic Differential Equations, Springer, 2000
- [2]. Ernest Ryu, Diffusion Models (Reverse-Time SDE), UCLA/SNU, 2024. (Online tutorial slides)
- [3] B. Anderson, Reverse-time diffusion equation models, Stochastic Processes and their Applications, 1982.
- [4] Tweedie's Formula ~ 1950 ; B. Efron, Tweedie's formula and Selection Bias, 2011
- [5] Special thanks to Litu Rout for extensive discussions.

1. SDE Basics

ODE: $dx_t = \mu(x_t, t) dt$, $x_t \in \mathbb{R}^d$
↳ drift term, x_0 given



Roughly: $x_{t+h} - x_t \approx \mu(x_t, t) \cdot h + o(h)$
↳ h small ↳ vanishingly small w.r.t h , for small h .

SDE: $dx_t = \mu(x_t, t) dt + \sigma(x_t, t) dB_t$
random variable ↳ drift term ↳ diffusion term ↳ Brownian motion

$x_0 \sim p_0(\cdot)$ (IC sampled from a given dist. $p_0(\cdot)$),
 $t \in [0, T]$ (or $t \in [0, \infty)$)

- B_t : Brownian motion
① $B_0 = 0$ a.s.
② $(B_t - B_s) \sim N(0, t-s)$
③ Independent increments.

What does this mean?



$$(X_{t+h} - X_t) \approx \mu(X_t, t) ((t+h) - t) + \sigma(X_t, t) (B_{t+h} - B_t) \\ + \text{higher order terms (depends on } ((t+h)-t)^2, (B_{t+h}-B_t)^2, \text{ etc.)}$$

$$\text{i.e. } X_{t+h} \approx X_t + \mu(X_t, t) \cdot h + \sigma(X_t, t) (B_{t+h} - B_t) \\ + \text{higher-order terms}$$

Important to note:

$$\textcircled{1} \quad ((t+h) - t)^2 = h^2 = o(h)$$

$$\textcircled{2} \quad f \text{ is smooth: } (f(t+h) - f(t))^2$$

$$= \left(f(t) + h \cdot f'(t) + \frac{h^2}{2} \cdot f''(t) + \frac{h^3}{6} f'''(\xi) - f(t) \right)^2$$

Suppose f has bounded derivative (say L).

$$\rightarrow \approx h^2 \cdot (f'(t))^2 \leq h^2 L^2 = o(h)$$

$$\textcircled{3} \quad (B(t+h) - B(t))^2 \approx ?$$

$$E[(B(t+h) - B(t))^2] = h \neq o(h)$$

$$E[B(t+h) - B(t)] = 0.$$

Takeaway: Second-order Taylor series expansion terms cannot be ignored for the stochastic setting.

Lipschitz continuity

Fact: Under "nice" conditions on $\mu(\cdot), \sigma(\cdot)$, $X_t, t \geq 0$ is well defined, has a unique t -continuous solution

Note: $dx_t = b(x_t)dt + \sigma(x_t)dB_t$ is intuitively Markovian, because to evolve $X_t \rightarrow X_{t+h}$, we only need to know X_t , and the evolution of B_t i.e., $B_t \rightarrow B_{t+h}$. Since B_t is indep increment, the past evolution of B_t does not affect its future.

Fact: Under nice conditions, $\{X_t, t \geq 0\}$ has the Strong Markov property, i.e.,

$$E_x [f(X_{\tau+h}) | \mathcal{F}_\tau] = E_{x_\tau} [f(X_h)]$$

where τ is a stopping time.

② How does the pdf of X_t evolve?

$\{X_t, t \geq 0\}$ a Markov process.

$P_t(\cdot) \equiv$ pdf of X_t ; $X_0 \sim p_0(\cdot)$

$\hookrightarrow$ pdf at time t .

$\nearrow \in \mathbb{R}^d$.
d-dim "marginal" $\downarrow$ at time $= 0$.

Answer: The Fokker-Planck Equation
 $\equiv$ Kolmogorov Forward Equation
 $\equiv$ "Master" equation.

(See also wikipedia for the history of this equation)

The FP equations is a PDE that describes the time evolution of $P_t(\cdot)$. Below, we assume existence, unique of solution (holds under "usual, nice" conditions. (See [1]).

$$\frac{\partial}{\partial t} p_t(x) = - \sum_{i=1}^d \frac{\partial}{\partial x_i} (\mu_i(x, t) p_t(x))$$

$$+ \sum_{i=1}^d \sum_{j=1}^d \left(D_{ij}(x, t) \cdot \frac{\partial^2}{\partial x_i \partial x_j} p_t(x) \right)$$

where $D_{ij}(x, t) = \frac{1}{2} \sum_{r=1}^d \sigma_{ir}(x, t) \cdot \sigma_{jr}(x, t).$

Note: The Fokker-Planck Equation is closely related to the continuity equation for ODEs of density flow. If we set $\sigma = 0$ in the above equation, we get

$$\frac{\partial p}{\partial t} + \nabla \cdot (\mu p) = 0, \text{ where}$$

$$p \equiv p_t(x) \quad ; \quad \mu = \begin{pmatrix} \mu_1(x, t) \\ \vdots \\ \mu_d(x, t) \end{pmatrix}$$

and divergence: $\nabla = \left(\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_d} \right)$

i.e., $\frac{\partial p}{\partial t} = - \sum_{i=1}^d \frac{\partial}{\partial x_i} (\mu_i(x, t) p_t(x)).$

If we start an ODE in a vector field $u(x, t)$, and with a random initial condition (IC), the density evolution is described by the continuity equation above.

This is also closely connected to the heat equation

$$\frac{\partial p}{\partial t} = \Delta p, \quad \text{i.e.,} \quad \frac{\partial p}{\partial t} = \sum_{i=1}^d \frac{\partial^2}{\partial x_i^2} p(t, x)$$

↳ Laplacian

Recall! $f = f(x_1, x_2, \dots, x_d)$ → Laplacian

$\nabla^2 f = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_d} \\ \vdots & \ddots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_d \partial x_1} & \dots & \dots & \frac{\partial^2 f}{\partial x_d^2} \end{pmatrix}$

↓
Hessian

$$\Delta f = \nabla \cdot \nabla f$$

$$= \sum_{i=1}^d \frac{\partial^2}{\partial x_i^2} f(x)$$

In the heat equation, $p(\cdot)$ is the temperature, and $\Delta p(x)$ encodes the average difference in temperature between x and the neighborhood of x . This equation describes the diffusion of heat over time.

The Fokker-Planck Equation has both a drift and diffusion term. Also, as we see in section 6 (pp. 29), we can absorb the diffusion term into the drift term via a time-varying vector field.

Sketch of proof of the FP Equation (for 1-d).

(Proof below adapted from [2]; see [1] for a complete proof).

Spcl case: $dx_t = \mu(x_t, t)dt + \sigma(x_t, t)dB_t$

$$\frac{\partial}{\partial t} P_t(z) = -\frac{\partial}{\partial z} (\mu(z, t) P_t(z)) + \frac{1}{2} \frac{\partial^2}{\partial z^2} \sigma^2(z, t) P_t(z)$$

Let $c(\cdot)$ be any bounded, differentiable function (assume existence of higher derivatives as needed).

Then, $\frac{\partial}{\partial t} E[c(x_t)]$

$$\stackrel{h \text{ small}}{\approx} \frac{1}{h} E[c(x_{t+h}) - c(x_t)]$$

from approx. of
SDE, (1), p. 5

$$\approx \frac{1}{h} E \left[c \left(x_t + \mu(x_t, t) \cdot h + \sigma(x_t, t) \cdot (B_{t+h} - B_t) \right) - c(x_t) \right]$$

Taylor's Thm: $c(x + \delta_1 + \delta_2) \approx c(x) + c'(x) \cdot \delta_1 + c'(x) \cdot \delta_2$
 $+ c''(x) \cdot \frac{\delta_1^2}{2} + c''(x) \cdot \frac{\delta_2^2}{2} + c''(x) \delta_1 \delta_2 + \text{cubic terms.}$

$$\begin{aligned}
 & \approx \frac{1}{h} E \left[\underbrace{c(x_t)}_{E[\cdot]=0} + \underbrace{c'(x_t) \mu(x_t, t) \cdot h}_{E[\cdot]=0} + \underbrace{c'(x_t) \sigma(x_t, t) (B_{t+h} - B_t)}_{E[\cdot]=h, \text{ indep increment}} \right. \\
 & \quad \left. + \underbrace{c'(x_t) h}_{E[\cdot]=0} + \underbrace{\frac{c''(x_t) \cdot \mu^2(x_t, t) h^2}{2}}_{d(h) \approx 0} + \underbrace{\frac{c''(x_t) \sigma^2(x_t, t) (B_{t+h} - B_t)^2}{2}}_{E[\cdot]=h} \right] \\
 & \Rightarrow E \left[P(x_t) (B_{t+h} - B_t)^2 \right] \\
 & = E[P(x_t)] \cdot h
 \end{aligned}$$

$$\approx \frac{1}{h} E \left[\cancel{c'(x_t) \mu(x_t, t) \cdot h} + \frac{c''(x_t) \sigma^2(x_t, t) \cdot h}{2} \right]$$

By tightening up and taking appropriate limit as $h \rightarrow 0$,

$$\frac{\partial}{\partial t} E[c(x_t)] = E \left[c'(x_t) \mu(x_t, t) + \frac{c''(x_t) \sigma^2(x_t, t)}{2} \right]$$

Expanding the expectations, we have

$$\frac{\partial}{\partial t} \int_{-\infty}^{\infty} c(z) p_t(z) dz =$$

$$\int_{-\infty}^{\infty} \left(c'(z) \mu(z, t) + c''(z) \frac{\sigma^2(z, t)}{2} \right) p_t(z) dz$$

Now, we have:

$$\textcircled{a} \quad \frac{\partial}{\partial t} \int_{-\infty}^{\infty} c(z) p_t(z) dz = \int_{-\infty}^{\infty} c(z) \frac{\partial}{\partial t} p_t(z) dz.$$

$$\textcircled{b} \quad \int_{-\infty}^{\infty} \underbrace{c'(z)}_{u'} \underbrace{\mu(z, t) p_t(z)}_v dz = \left[\underbrace{c(z)}_u \underbrace{\mu(z, t) p_t(z)}_v \right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \underbrace{c(z)}_u \cdot \frac{\partial}{\partial z} \underbrace{(\mu(z, t) p_t(z))}_v dz.$$

Integration by parts

$$\int_a^b u(z) v'(z) dz = u(z) v(z) \Big|_a^b - \int_a^b u'(z) v(z) dz.$$

Now assume random variables are proper $\forall t$: Then

$$p_t(\infty) = p_t(-\infty) = 0 \quad \forall t.$$

$$\therefore c(z) \mu(z, t) p_t(z) \Big|_{-\infty}^{\infty} = 0.$$

$$\Rightarrow \int_{-\infty}^{\infty} c'(z) u(z, t) p_t(z) dz = \int_{-\infty}^{\infty} c(z) \left(-\frac{\partial}{\partial z} (u(z, t) p_t(z)) \right) dz$$

$$\begin{aligned} \textcircled{c} \quad \frac{1}{2} \int_{-\infty}^{\infty} \underbrace{c'(z)}_{v'} \underbrace{\sigma^2(z, t)}_u p_t(z) dz &= \frac{1}{2} \sigma^2(z, t) p_t(z) \cdot \underbrace{c'(z)}_v \Big|_{-\infty}^{\infty} \xrightarrow{=0 \text{ at}} \\ &- \frac{1}{2} \int_{-\infty}^{\infty} \underbrace{c'(z)}_v \cdot \underbrace{\frac{\partial}{\partial z} (\sigma^2(z, t) p_t(z))}_{u'} dz \end{aligned}$$

Next,

$$\int_{-\infty}^{\infty} \underbrace{c'(z)}_{v'} \underbrace{\frac{\partial}{\partial z} (\sigma^2(z, t) p_t(z))}_u dz$$

$$= c(z) \cdot \frac{\partial}{\partial z} (\sigma^2(z, t) p_t(z)) \Big|_{-\infty}^{\infty}$$

$$- \int_{-\infty}^{\infty} c(z) \frac{\partial^2}{\partial z^2} (\sigma^2(z, t) p_t(z)) dz.$$

Assume further: $\frac{\partial}{\partial z} \sigma^2(z, t) p_t(z) = 0$ at $z = -\infty, \infty$

essentially means that $\frac{\partial}{\partial z} p_t(-\infty) = \frac{\partial}{\partial z} p_t(\infty) = 0$

(a stronger condition than properness, but is reasonable)

$$\begin{aligned}\therefore \frac{1}{2} \int_{-\infty}^{\infty} c'(z) \sigma^2(z, t) p_t(z) dz \\ = \frac{1}{2} \int_{-\infty}^{\infty} c(z) \frac{\partial^2}{\partial z^2} (\sigma^2(z, t) p_t(z)) dz\end{aligned}$$

Substituting (a), (b), (c) in (x),

$$\begin{aligned}\int_{-\infty}^{\infty} c(z) \left(\frac{\partial}{\partial t} p_t(z) \right) dz &= \int_{-\infty}^{\infty} c(z) \left(-\frac{\partial}{\partial z} (\mu(z, t) p_t(z)) \right) dz \\ &+ \int_{-\infty}^{\infty} c(z) \left(\frac{1}{2} \frac{\partial^2}{\partial z^2} (\sigma^2(z, t) p_t(z)) \right) dz\end{aligned}$$

Since above is true for all "reasonable" $c(z)$

we have

$$\frac{\partial}{\partial t} p_t(z) = -\frac{\partial}{\partial z} (\mu(z, t) p_t(z)) + \frac{1}{2} \frac{\partial^2}{\partial z^2} (\sigma^2(z, t) p_t(z))$$

This is the 1-d FP Equation



Aside:

Note also that the formal proof is different, and uses Dynkin's formula using Ito calculus

Theorem 7.4.1 (Dynkin's formula).

Let $f \in C_0^2(\mathbf{R}^n)$. Suppose τ is a stopping time, $E^x[\tau] < \infty$. Then

$$E^x[f(X_\tau)] = f(x) + E^x\left[\int_0^\tau Af(X_s)ds\right]. \quad (7.4.1)$$

← Oksendal, Chap 7j

A is the generator of the Markov Process

See Oksendal, prob. 8.3 for details.

8.3. (Kolmogorov's forward equation)

(aka Fokker-Planck Equation)

Let X_t be an Itô diffusion in $\mathbf{R}^n$ with generator

$$Af(y) = \sum_{i,j} a_{ij}(y) \frac{\partial^2 f}{\partial y_i \partial y_j} + \sum_i b_i(y) \frac{\partial f}{\partial y_i}; \quad f \in C_0^2$$

and assume that the transition measure of X_t has a density $p_t(x, y)$, i.e. that

$$E^x[f(X_t)] = \int_{\mathbf{R}^n} f(y) p_t(x, y) dy; \quad f \in C_0^2. \quad (8.6.34)$$

Assume that $y \mapsto p_t(x, y)$ is smooth for each t, x . Prove that $p_t(x, y)$ satisfies the Kolmogorov forward equation

$$\frac{d}{dt} p_t(x, y) = A_y^* p_t(x, y) \quad \text{for all } x, y, \quad (8.6.35)$$

where A_y^* operates on the variable y and is given by

$$A_y^* \phi(y) = \sum_{i,j} \frac{\partial^2}{\partial y_i \partial y_j} (a_{ij} \phi) - \sum_i \frac{\partial}{\partial y_i} (b_i \phi); \quad \phi \in C^2 \quad (8.6.36)$$

(i.e. A_y^* is the adjoint of A_y .)

(Hint: By (8.6.34) and Dynkin's formula we have

$$\int_{\mathbf{R}^n} f(y) p_t(x, y) dy = f(x) + \int_0^t \int_{\mathbf{R}^n} A_y f(y) p_s(x, y) dy ds; \quad f \in C_0^2.$$

Now differentiate w.r.t. t and use that

$$\langle A\phi, \psi \rangle = \langle \phi, A^* \psi \rangle \quad \text{for } \phi \in C_0^2, \psi \in C^2,$$

where $\langle \cdot, \cdot \rangle$ denotes inner product in $L^2(dy)$.)

(Note: The Dynkin's formula is the analog of the fundamental theorem of calculus for diffusions.)

③ The Ornstein-Uhlenbeck (OU) Process.

$$dX_t = -X_t dt + \sqrt{2} dB_t, \quad X_t \in \mathbb{R}^d$$

$p_t(\cdot)$ is the density at time t ; $X_0 \sim p_0(\cdot)$

The FP Equation (for $d=1$).

$$\frac{\partial}{\partial t} p_t(x) = \frac{\partial}{\partial x} (x p_t(x)) + \frac{\partial^2}{\partial x^2} p_t(x),$$

Solution: (in steady state as $T \rightarrow \infty$).

$$\frac{\partial}{\partial t} p_\infty(x) = 0 = x \frac{\partial}{\partial x} p_\infty(x) + p_\infty(x) + \frac{\partial^2}{\partial x^2} p_\infty(x)$$

Claim: $p_\infty(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, i.e., $N(0, 1)$

Pf: $\frac{\partial}{\partial x} p_\infty(x) = -x \frac{1}{\sqrt{2\pi}} e^{-x^2/2} = -x p_\infty(x)$

$$\frac{\partial^2}{\partial x^2} p_\infty(x) = -p_\infty(x) + x^2 p_\infty(x)$$

$$\therefore \frac{\partial^2}{\partial x^2} p_{\infty}(x) + x \frac{\partial}{\partial x} p_{\infty}(x) + p_{\infty}(x)$$

$$= -p_{\infty}(x) + x^2 p_{\infty}(x) + x \cdot (-x p_{\infty}(x)) + p_{\infty}(x)$$

$$= 0.$$



Takeaway: Start the OU process with $X_0 \sim p_0(\cdot)$, any pdf. The stationary dist. is a Gaussian $\sim N(0, 1)$.

④ Reversing the OU Process

We will denote the OU process as the "forward process", because it starts at $t=0$ with $X_0 \sim p_0(\cdot)$ (the distribution over images), and as time evolves, $p_t(x) \rightarrow N(0, 1)$.

In this part, we write the OU process as:

$$d\vec{X}_t = -\vec{X}_t dt + \sqrt{2} dB_t, \quad X_0 \sim \vec{P}_0(\cdot), \\ t \in [0, T].$$

We denote the pdf of $\vec{X}_t \sim \vec{P}_t(\cdot)$.

Our goal is to construct a stochastic process that runs backwards in time, i.e., $t \in [T, 0]$,

s.t. $\overleftarrow{X}_T \sim \vec{P}_T$, i.e., $\overleftarrow{X}_T \stackrel{d}{=} \vec{X}_T$, and also $\overleftarrow{X}_0 \stackrel{d}{=} \vec{X}_0$.

In other words, this process is initialized with a sample from $\vec{P}_T(\cdot)$ (the terminal state of the forward OU process, which is roughly a standard Gaussian, for T large enough).

After running this new "reverse process", $\overleftarrow{X}_t$, $t \in [T, 0]$, at time '0', we will get a sample from $\vec{P}_0(\cdot)$, i.e., the distribution over images.

$$\text{Let : } d\overleftarrow{X}_t = \left(-\overleftarrow{X}_t - 2\nabla_{\vec{x}} \ln \vec{P}_t(\vec{x}) \right) dt \\ + \sqrt{2} d\tilde{B}_t,$$

$$t \in [T, 0], \quad \overleftarrow{X}_T \sim \vec{P}_T(\cdot).$$

Denote the pdf of this reverse process by $\overleftarrow{P}_t(\cdot)$, $t \in [T, 0]$. By construction, we have $\overleftarrow{P}_T(\cdot) = \overrightarrow{P}_T(\cdot)$.

Claim: $\forall t \in [0, T]$, $\overleftarrow{P}_t(\cdot) = \overrightarrow{P}_t(\cdot)$.

The above claim was derived in [3] (Anderson 1982).

The proof of the claim below follows the discussion in [2]. (Anderson derives the reverse SDE for any general forward process $d\overrightarrow{X}_t = \mu(X_t)dt + \sigma(X_t)d\tilde{B}_t$, see [3] for details.)

Proof: Consider the reverse SDE:

$$d\overleftarrow{X}_t = \left(-\overleftarrow{X}_t - 2 \nabla \ln \overrightarrow{P}_t(\overleftarrow{X}_t) \right) dt + \sqrt{2} d\tilde{B}_t$$

$t \in [T, 0]$, $\overleftarrow{X}_T \sim \overrightarrow{P}_T(\cdot)$.

$t \rightarrow T-t$: (time direction reverse).

$$d\overleftarrow{X}_{T-t} = -\overleftarrow{X}_{T-t} - 2 \nabla \ln \overrightarrow{P}_{T-t}(\overleftarrow{X}_{T-t}) d(T-t)$$

$$+ \sqrt{2} d\tilde{B}_{\tau-t}.$$

$$\text{Now } d(\tau-t) = -dt \quad ; \quad \tilde{B}_{\tau-t} \stackrel{d}{=} \hat{B}_t$$

$$\therefore d\overleftarrow{X}_{\tau-t} = \left(\overleftarrow{X}_{\tau-t} + 2\nabla \ln \vec{p}_{\tau-t}(\overleftarrow{X}_{\tau-t}) \right) dt + \sqrt{2} d\hat{B}_t.$$

$$\text{Let } Y_t = \overleftarrow{X}_{\tau-t}$$

$$\text{Then, } dY_t = (Y_t + 2\nabla \ln \vec{p}_{\tau-t}(Y_t))dt + \sqrt{2} d\hat{B}_t, \\ t \in [0, \tau], \quad Y_0 \sim \vec{p}_{\tau}(\cdot).$$

Let $q_t(\cdot)$ be the pdf of Y_t , with $q_0(\cdot) = \vec{p}_{\tau}(\cdot)$.

The FP Equation for Y_t : $\left(\nabla_x(\cdot) = \frac{\partial}{\partial x}(\cdot) \right)$

$$\frac{\partial}{\partial t} q_t(y) = -\frac{\partial}{\partial y} \left((y + 2\frac{\partial \ln \vec{p}_{\tau-t}}{\partial y}(y)) q_t(y) \right) \\ + \frac{\partial^2}{\partial y^2} q_t(y).$$

Let $\overleftarrow{p}_t(\cdot)$ be the pdf of $\overleftarrow{X}_t$. Then, by

construction, $\overleftarrow{p}_t(\cdot) = q_{T-t}(\cdot)$.

$\therefore$ The FP Equation for $\overleftarrow{p}_t(\cdot)$ is given by:

$$\frac{\partial}{\partial t} \overleftarrow{p}_{T-t}(x) = -\frac{\partial}{\partial x} \left(\left(x + 2 \frac{\partial \ln \overrightarrow{p}_{T-t}(x)}{\partial x} \right) \cdot \overleftarrow{p}_{T-t}(x) \right) + \frac{\partial^2}{\partial x^2} \overleftarrow{p}_{T-t}(x).$$

$(T-t) \rightarrow s :$

$$-\frac{\partial}{\partial s} \overleftarrow{p}_s(x) = -\frac{\partial}{\partial x} \left(\left(x + 2 \frac{\partial \ln \overrightarrow{p}_s(x)}{\partial x} \right) \overleftarrow{p}_s(x) \right) + \frac{\partial^2}{\partial x^2} \overleftarrow{p}_s(x)$$

(i.e.,

$$\frac{\partial}{\partial s} \overleftarrow{p}_s(x) = \frac{\partial}{\partial x} (x \overleftarrow{p}_s(x)) + \frac{\partial}{\partial x} \left(2 \cdot \frac{1}{\overrightarrow{p}_s(x)} \frac{\partial}{\partial x} \overrightarrow{p}_s(x) \right) \overleftarrow{p}_s(x) - \frac{\partial^2}{\partial x^2} \overleftarrow{p}_s(x) \quad \longrightarrow \textcircled{3}$$

We now claim that $\overleftarrow{p}_s(x) = \overrightarrow{p}_s(x)$. To show this, we substitute $\overrightarrow{p}_s(x) = \overleftarrow{p}_s(x)$ in $\textcircled{3}$ above:

$$\frac{\partial}{\partial s} \hat{p}_s(x) = \frac{\partial}{\partial x} (x \hat{p}_s(x)) + 2 \cdot \frac{\partial^2}{\partial x^2} \hat{p}_s(x) - \frac{\partial^2}{\partial x^2} \hat{p}_s(x)$$

i.e.,

$$\frac{\partial}{\partial s} \hat{p}_s(x) = \frac{\partial}{\partial x} (x \hat{p}_s(x)) + \frac{\partial^2}{\partial x^2} \hat{p}_s(x) \rightarrow \textcircled{4}$$

Recall the FP Equation for the standard OU Process $\vec{X}(t)$:

$$\frac{\partial}{\partial t} \vec{p}_t(x) = \frac{\partial}{\partial x} (x \vec{p}_t(x)) + \frac{\partial^2}{\partial x^2} \vec{p}_t(x),$$

$\rightarrow \textcircled{5}$

Observe that $\textcircled{4} = \textcircled{5}$. From uniqueness of solution of the FP Equation, we conclude that our claim that $\hat{p}_s(x) = \vec{p}_s(x)$ is correct. $\square$

Note: $\nabla \ln \vec{p}_t(x)$ is called the **Score function** in the generative modeling literature. Also, see (Song & Ermon, 2019) for applying this approach for image generation.

⑤ Tweedie's Formula to compute $\nabla \ln p_t(x)$

The reverse diffusion starts with a Gaussian sample at $t=T$, and approximates a sample from the image distribution $p_0(\cdot)$ at $t=0$. This evolution, however, requires access to the score function $\nabla \ln p_t(\cdot)$.

There are several approaches to learning this function. One line of work (Hyvarinen 2005) develops an approach to learn this function by matching to minimize a Fisher information metric. This requires estimates of higher order derivatives, and in sliced score matching (Song et al. 2019) develop an approach using projections to get around higher order derivatives.

The most popular approach today is based on denoising score matching, proposed by (Vincent 2010). This technique goes back to work by (Tweedie 1950) as discussed by (Efron 2011).

Outline of this approach:

- ① Suppose we have access to a large collection of samples of images. Denote the r.v. corresponding to an image by $X_0 \sim p_0(\cdot)$.

We "noise" the image by adding '0' mean Gaussian noise, with appropriate variance σ^2 s.t.

$$X_\sigma = X_0 + \sigma Z, \quad Z \sim N(0, I)$$

Z indep. of X_0 .

and X_σ has the marginal pdf given by the forward OU process

- ② We now "denoise" the image using an MMSE estimator:

$$\hat{X}_0 = E[X_0 | X_\sigma = x_\sigma]$$

$$= g_\theta(x_\sigma)$$

parameters
of a neural network

(learned
function).

This estimator is learned using squared-loss regression:

$$\min_{g_\theta(\cdot)} E[(X_0 - g_\theta(X_\sigma))^2]$$

If $g_\theta(\cdot)$ is a large enough class of functions, we know that

$$g_\theta(x_\sigma) = E[X_0 | X_\sigma = x_\sigma].$$

The optimization above is solved using a large number of (image, noised image) pairs, i.e., (X_0, X_σ) pairs.

③ Tweedie's Formula: This goes back to 1950s, and states: Suppose $X_\sigma = X_0 + \sigma Z$ as above, then,

$\searrow \sim p_0 \quad \searrow \sim N(0,1)$

$$E[X_0 | X_\sigma = x_\sigma] = x_\sigma + \sigma^2 \nabla \ln p_\sigma(x_\sigma).$$

Thus, by learning the LHS, we effectively learn a score function $\nabla \ln p_t(\cdot)$.

(The actual implementations differ from above pipeline, but is based on the same idea).

Proof of Tweedie's formula (for scalar case; $\sigma=1$)

Suppose $Y = X + Z$, $X \sim p(\cdot)$, $X \perp\!\!\!\perp Z$

$Z \sim N(0,1)$ $\downarrow$ indep

$$Y \sim q(\cdot) = p(x) * \left(\frac{1}{\sqrt{2\pi}} e^{-z^2/2} \right)$$

$$E[X|Y=y] = \int_{-\infty}^{\infty} f_{X|Y}(x|y) \cdot x \, dx$$

$$= \int_{-\infty}^{\infty} \frac{f_{X,Y}(x,y)}{f_Y(y)} \cdot x \, dx$$

$$= \frac{1}{f_Y(y)} \int_{-\infty}^{\infty} x f_{Y|X}(y|x) \cdot f_X(x) dx$$

Now, $f_Y(y) = q(y)$, $f_X(x) = p(x)$

$$f_{Y|X}(y|x) = \frac{1}{\sqrt{2\pi}} e^{-(y-x)^2/2}$$

$$\therefore E[X|Y=y] = \frac{1}{q(y)} \int_{-\infty}^{\infty} x \cdot p(x) \cdot \frac{e^{-(y-x)^2/2}}{\sqrt{2\pi}} dx$$

↪ (1)

Next, we have:

$$q(y) = p(y) * \frac{1}{\sqrt{2\pi}} e^{-y^2/2}$$

$$= \int_{-\infty}^{\infty} p(x) \cdot \frac{1}{\sqrt{2\pi}} e^{-(y-x)^2/2} dx$$

$$\therefore \frac{\partial}{\partial y} q(y) = \int_{-\infty}^{\infty} p(x) \cdot \frac{1}{\sqrt{2\pi}} e^{-(y-x)^2/2} \cdot \left(-2 \frac{(y-x)}{2}\right) dx$$

$$= -y \int_{-\infty}^{\infty} \underbrace{p(x)}_{f_X(x)} \cdot \underbrace{\frac{1}{\sqrt{2\pi}} e^{-(y-x)^2/2}}_{f_{Y|X}(y|x)} dx$$

$$+ \int_{-\infty}^{\infty} x p(x) \cdot \frac{1}{\sqrt{2\pi}} e^{-(x-y)^2/2} dx$$

$$q(y) \cdot E[X|Y=y]$$

from (1')

$$= -y \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx + q(y) E[X|Y=y]$$

$$= -y q(y) + q(y) E[X|Y=y]$$

$$\therefore \frac{\partial}{\partial y} q(y) = -y q(y) + q(y) E[X|Y=y]$$

→ (2')

$$\text{i.e., } E[X|Y=y] = y + \frac{1}{q(y)} \cdot \frac{\partial}{\partial y} q(y) \\ = y + \frac{\partial}{\partial y} \ln q(y). \quad \square$$

⑥ Generative Flow Models.

Refs: ① Song et. al., Score-based generative modeling through stochastic differential equations, arXiv 2011.13456, 2020.

② Song et. al., Denoising diffusion implicit models, arXiv: 2010.02502, 2020.

High level idea so far: Construct an SDE running backward in time to sample from the distribution over images.

This section: Construct an ODE, initialized with a RANDOM initial condition, so that the terminal state is an image. We do this through a 2-step process:

① Construct a time-dependent vector field $\vec{V}(x, t)$, $x \in \mathbb{R}^d$, such that:

$$d\vec{Z}_t = \vec{V}(\vec{Z}_t, t) dt, \quad \vec{Z}_0 \sim p_0(\cdot).$$

i.e., sample an image $\vec{Z}_0$ from $p_0(\cdot)$, and evolve the deterministic dynamic forward in time.

The goal is to construct $\vec{V}(\cdot, \cdot)$ such that $\vec{Z}_\infty \rightarrow N(0, I)$. (i.e., the ODE flows to a sample of a Gaussian r.v.; note that because the IC is a random variable, $\vec{Z}_t$ is a r.v. $\forall t \geq 0$).

② Reverse the Field: Construct a new reverse vector field $\overleftarrow{V}(z, t)$ such that it is initialized as: $\overleftarrow{Z}_0 = \vec{Z}_T$. This ODE also runs forwards in time, and the field $\overleftarrow{V}(\cdot, \cdot)$ is constructed such that $\overleftarrow{Z}_T = \vec{Z}_0$, i.e., you sample an image.

Note: This approach is much faster to implement by using fast ODE solvers.

Key Idea: Interpreting an SDE density evolution as an ODE density evolution:

Consider: $dx_t = \mu(x_t, t)dt + \sigma(x_t, t)dB_t$,
 $t \in [0, T]$, $x_t \in \mathbb{R}$ (scalar case for simple notation)

The FP Equation is: ($P_t(x)$ is the density)

$$\frac{\partial}{\partial t} P_t(x) = -\frac{\partial}{\partial x} (\mu(x, t) P_t(x)) + \frac{1}{2} \frac{\partial^2}{\partial x^2} (\sigma^2(x, t) P_t(x)).$$

Observe:

$$\begin{aligned} \frac{1}{2} \frac{\partial^2}{\partial x^2} (\sigma^2 P_t) &= \frac{1}{2} \frac{\partial}{\partial x} \left(\frac{\partial}{\partial x} \sigma^2 P_t \right) \\ &= \frac{1}{2} \frac{\partial}{\partial x} \left(\left(\frac{\partial}{\partial x} \sigma^2 \right) P_t + \sigma^2 \cdot \frac{\partial}{\partial x} P_t \right) \\ &\stackrel{\left(\because \frac{\partial}{\partial x} \ln P_t = \frac{1}{P_t} \frac{\partial}{\partial x} P_t \right)}{=} \frac{1}{2} \frac{\partial}{\partial x} \left(\left(\frac{\partial}{\partial x} \sigma^2 \right) P_t + \sigma^2 \cdot \left(\frac{\partial}{\partial x} \ln P_t \right) P_t \right) \end{aligned}$$

We can now absorb the second term in the FP Equation into the first.

$$\frac{\partial}{\partial t} P_t(x) = - \frac{\partial}{\partial x} \left(\underbrace{\left(\mu(x,t) - \frac{1}{2} \frac{\partial}{\partial x} \sigma^2(x,t) - \frac{\sigma^2(x,t)}{2} \frac{\partial}{\partial x} \ln P_t(x) \right)}_{\vec{V}(x,t)} P_t(x) \right)$$

$$= - \frac{\partial}{\partial x} \left(\vec{V}(x,t) P_t(x) \right).$$

This is the density flow equation for an ODE in a vector field $\vec{V}(x,t)$, with initial condition $P_0(x)$.

(Why: In the FP equation, set $\sigma(\cdot) = 0$, and we get the continuity equation, which describes the density flow for an ODE; see page 8).

① Using reasoning as above, define

$$\vec{V}(x, t) = -\left(x + \nabla \ln p_t(x)\right), \quad x \in \mathbb{R}^d.$$

where $p_t(\cdot)$ is the pdf of the OU process:

$$d\vec{X}_t = -\vec{X}_t dt + \sqrt{2} dB_t, \quad X_0 \sim p_0(\cdot).$$

Then, consider the ODE

$$d\vec{Z}_t = \vec{V}(\vec{Z}_t, t) dt, \quad Z_0 \sim p_0(\cdot).$$

Then, $\vec{Z}_t \stackrel{d}{=} \vec{X}_t$.

i.e., the ODE has the same marginal as the SDE.

② Reversing the field

Define $\overleftarrow{V}(x, t) = -\vec{V}(x, T-t)$, $t \in [0, T]$
as the reverse vector field, and let

$$d\overleftarrow{\mathbf{z}}_t = \overleftarrow{\mathbf{v}}(\overleftarrow{\mathbf{z}}_t, t) dt, \quad t \in [0, \tau]$$

be the reverse field ODE.

Claim: Suppose $\overleftarrow{\mathbf{z}}_0 = \overrightarrow{\mathbf{z}}_\tau$ (initialization of reverse field)

Then, $\overleftarrow{\mathbf{z}}_\tau = \overrightarrow{\mathbf{z}}_0$. In words, this ODE flow samples from the image distribution.

Pf: $\frac{d\overleftarrow{\mathbf{z}}_t}{dt} = \overleftarrow{\mathbf{v}}(\overleftarrow{\mathbf{z}}_t, t) = -\overrightarrow{\mathbf{v}}(\overleftarrow{\mathbf{z}}_t, \tau - t).$

Change: $\tau - t \rightarrow s$ $\frac{d\overleftarrow{\mathbf{z}}_{\tau-s}}{ds} = \overrightarrow{\mathbf{v}}(\overleftarrow{\mathbf{z}}_{\tau-s}, s)$

Also, $\frac{d\overrightarrow{\mathbf{z}}_s}{ds} = \overrightarrow{\mathbf{v}}(\overrightarrow{\mathbf{z}}_s, s)$

i.e., $\overleftarrow{\mathbf{z}}_{\tau-s} = \overrightarrow{\mathbf{z}}_s \quad \forall s. \quad \square$

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